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Plasmon enhanced light-mediated chemical processes

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Abstract

In triplet-triplet annihilation photon upconversion (TTA-UC), two low-energy photons can be combined into a higher-energy photon by a weak light source in place of more energy-intensive methods. Our project focuses on optimizing the dielectric and acceptor layers of organic light-emitting diodes (OLEDs) to make TTA-UC more efficient in these devices. To study this, we evaporated organic thin films of varying thicknesses onto OLEDs and characterized these devices using electroluminescence (EL) and external quantum efficiency (EQE) measurements. Our final structure was a non-doped annihilator layer sandwiched between two sensitizer layers and no longer had a dielectric layer.

Objectives

- Improve the efficiency of triplet-triplet annihilation photon upconversion (TTA-UC) in solid-state systems
- Study TTA-UC through organic light-emitting diodes (OLEDs)
- Optimize the dielectric and acceptor layers of OLEDs

Impact

- New modifications that can improve the efficiency of TTA-UC with application to OLEDs
- New insight on optimizing TTA-UC in solid-state systems
- Future work can explore applicability of these results to other solid-state applications of TTA-UC

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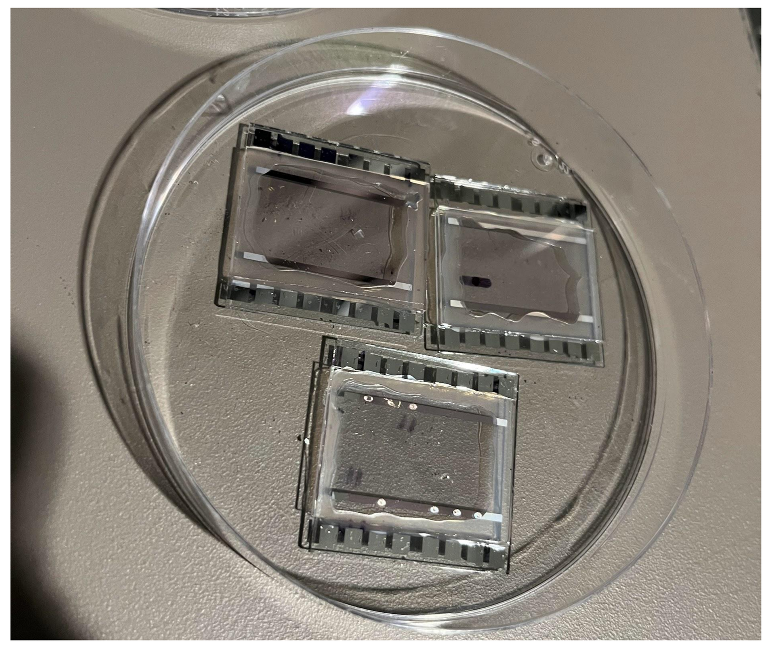
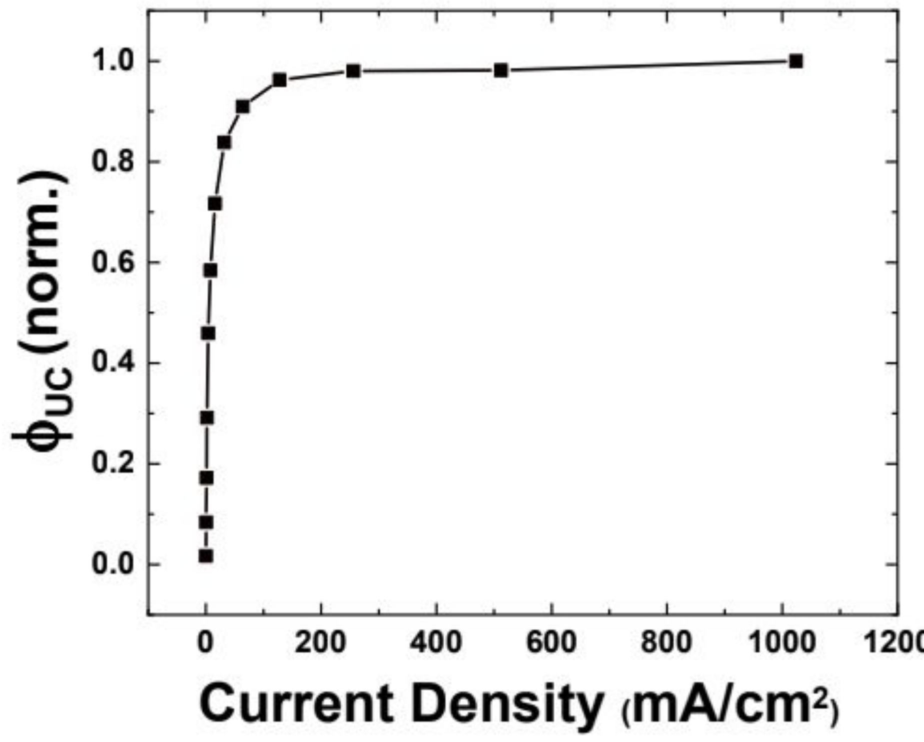
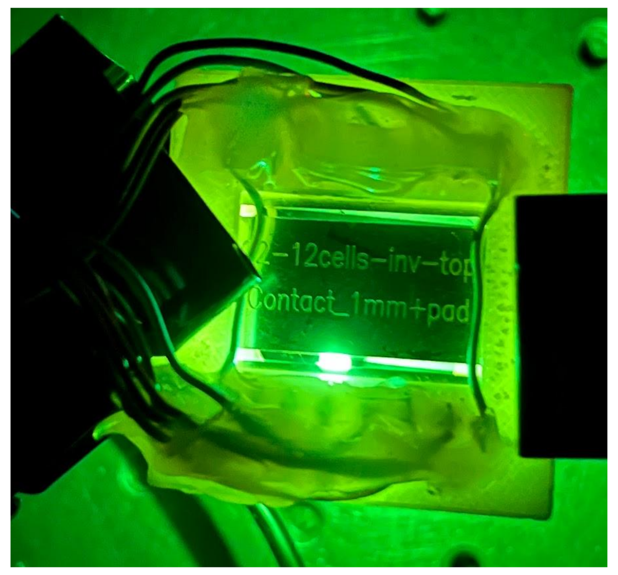
Methodology

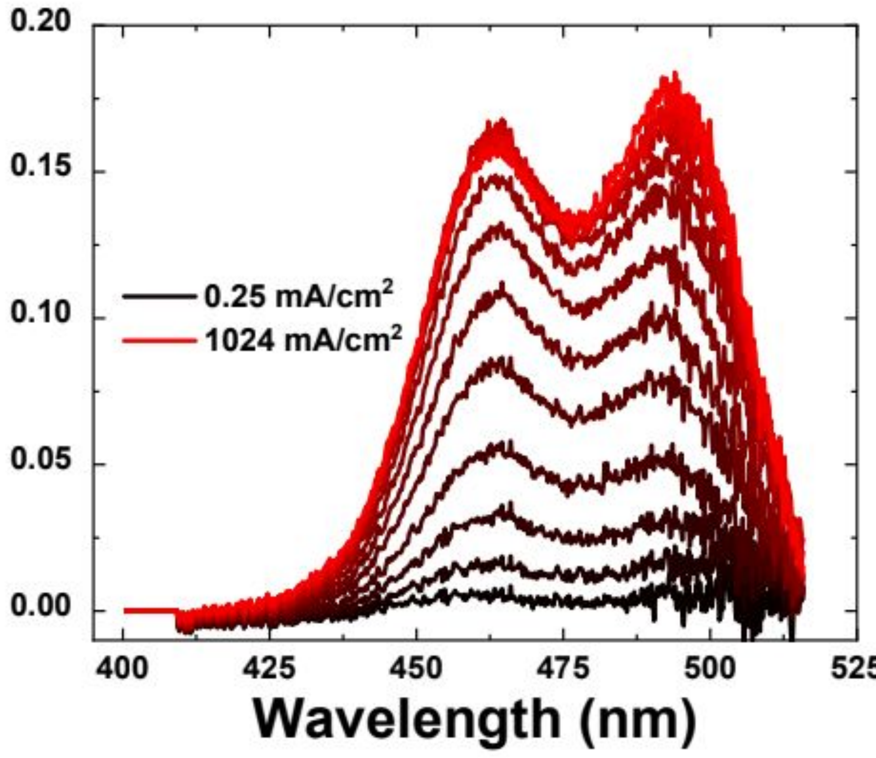
- Fabricate varying thicknesses of thin films using thermal evaporation
- Characterize devices using electroluminescence (EL) and external quantum efficiency (EQE) measurements
- Make adjustments to device recipe based on trends in data

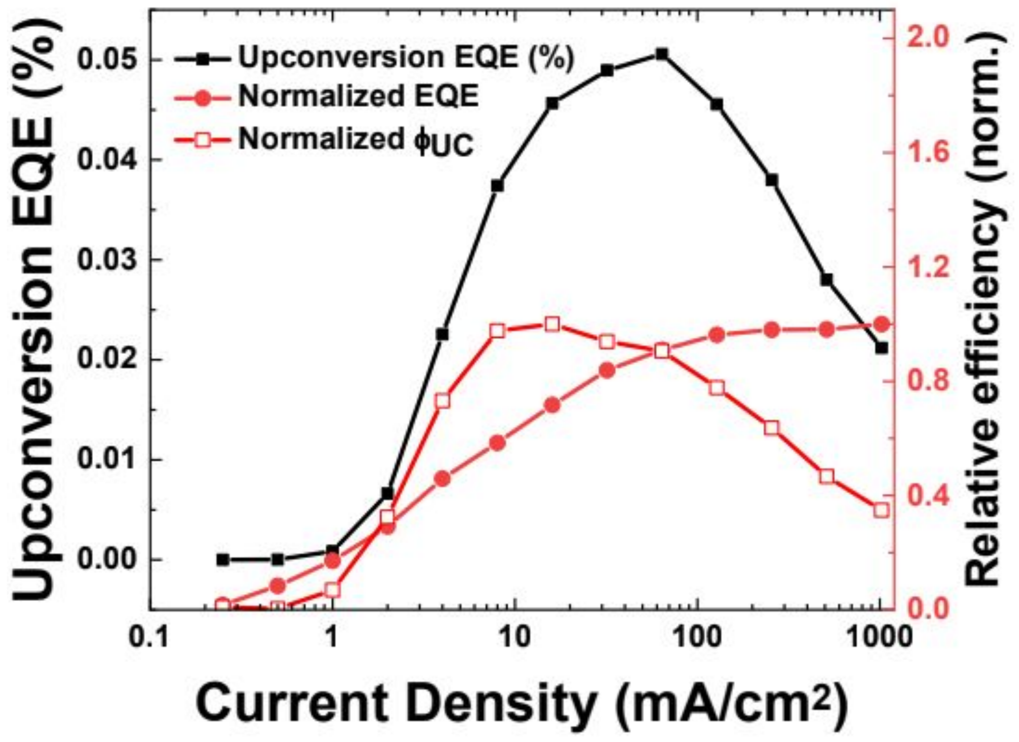




Results







- Sensitizer-annihilator-sensitizer device structure can maximize formation of singlet excitons
- Dielectric is unnecessary and may block the formation of singlet excitons
- Pure 2-methyl-9,10-bis(naphthalene-2-yl)anthracene (MADN) does not lead to significant concentration quenching and may not need to be doped with 4,4'-bis[4-(diphenylamino)styryl]biphenyl (BDAVBi)
- Final structure: 0nm TeO_2/ 10nm DMPPP: 40% PtOEP/ 40nm MADN/ 10nm DMPPP: 40% PtOEP.